EES 102: Introduction to Environmental Sciences

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"IT ALL STARTED WITH A BIG BANG…"

A Belgian priest named Georges Lemaître first suggested the big bang theory in the 1920s when he theorized that the universe began from a single primordial atom.

He derived the expansion of the Universe from the equations of general relativity in an 1927 article and proposed a first value for the Hubbles constant.

However many established scientists ridiculed the idea till the cosmic background radiation was discovered in the 1960s

The name was meant to be a joke by critics of the theory but it stuck

"IT ALL STARTED WITH A BIG BANG…"

The idea subsequently received major boosts by Edwin Hubble's observations that galaxies are speeding away from us in all directions.

In 1929, from analysis of galactic redshifts, Edwin Hubble concluded that galaxies are drifting apart; this is important observational evidence consistent with the hypothesis of an expanding universe.

Hubble found that the magnitude of the redshift is largest of galaxies furthest away from the Milky Way (our galaxy)

THE BIG BANG

The theory of our universe's origin centers on a cosmic cataclysm unmatched in all of history—the big bang. This theory was born of the observation that other galaxies are moving away from our own at great speed, in all directions, as if they had all been propelled by an ancient explosive force.

Before the big bang, scientists believe, the entire vastness of the observable universe, including all of its matter and radiation, was compressed into a hot, dense mass just a few millimeters across. This nearly incomprehensible state is theorized to have existed for just a fraction of the first second of time.

~13.8 billion years ago, a massive blast allowed all the universe's known matter and energy—even space and time themselves—to spring from some ancient and unknown type of energy.

The theory maintains that, in the instant—a trillion-trillionth of a second—after the big bang, the universe expanded with incomprehensible speed from its pebble-size origin to astronomical scope. Expansion has apparently continued, but much more slowly, over the ensuing billions of years.

STAGES OF THE BIG BANG

1. Superfast inflation from size of atom to size of a grapefruit in a fraction of a second – the big bang marks the beginning of space-time
2. Post inflation the universe is a seething hot soup of electrons, quarks and other particles – nowadays high energy physics studies these with particle colliders
3. A rapid cooling permits quarks to clump into protons and neutrons
4. At 3 minutes after the Big Bang the Universe is still too hot to for atoms to form. Charged electrons and protons prevent light from shining. The universe s a superhot fog. The remnants of the fog are seen I the cosmic microwave background radiation
5. At 300,000 years after the Big Bang the Universe has cooled enough for electrons to combine with protons and neutrons to form atoms. Mostly hydrogen and helium, light can finally shine.

STAGES OF THE BIG BANG

6. Gravity makes hydrogen and helium gas coalesce to form the giant clouds that will become galaxies. Smaller clumps of gas collapse onto themselves to form the first stars. The goldilocks conditions for this to happen were tiny variations in the distribution of matter and energy throughout the universe that can be seen in the maps of the cosmic background radiation
7. As galaxies cluster together under gravity the first stars die an spew heavy elements into space. Those will eventually turn into new stars and planets. Stars can only make elements up to iron by fusion. All heavier elements are made by neutron capture primarily in the blast that throws out elements, radioactive radiation and neutrons after supernova explosions.

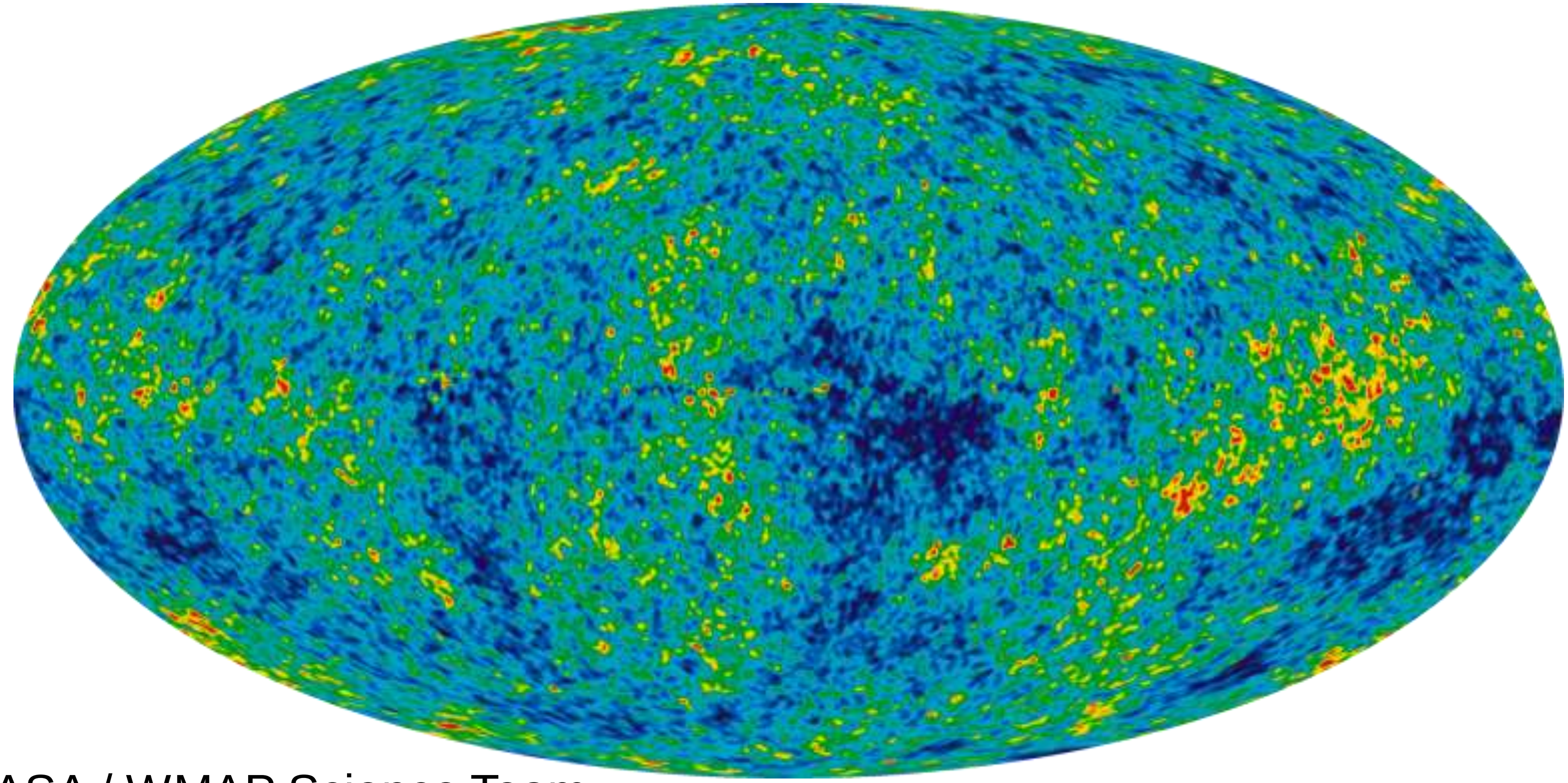
"IT ALL STARTED WITH A BIG BANG…"

The discovery of cosmic microwave radiation in 1965 by Arno Penzias and Robert Wilson, finally sealed the "Big Bang Theory".

The glow of cosmic microwave background radiation, which is found throughout the universe, is thought to be a tangible remnant of leftover light of the fog from after the big bang.

The radiation is akin to that used to transmit TV signals via antennas. But it is the oldest radiation known and may hold many secrets about the universe's earliest moments.

TEMPERATURE FLUCTUATIONS IN THE COSMIC BACKGROUND RADIATION



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HOW DO WE KNOW HOW OLD THE UNIVERSE IS?

Detailed measurements of the expansion rate of the universe place the Big Bang at around 13.8 billion years ago, which is thus considered the age of the universe.

Primer: When it is in red font and something you don't know already, it might be a good idea to mug it up before the exam!

WHERE DO ALL THE ELEMENTS COME FROM?

In the years following the Big Bang, hydrogen atoms floated freely around the Universe. These atoms were slightly more packed together in some places than in others.

In the more crowded areas, the hydrogen atoms were close enough to each other to let gravity do its work. Gravity pulled atoms in area with more matter closer together. As atoms were squashed closer and closer together, they banged into each other more and more violently and the regions where that happened heated up. When the cloud got so hot that protons & electrons split apart, a plasma (>3000 K) was recreated. Once temperature reached 10 Million K the collisions became so violent that nuclei fused together to create new elements and huge amounts of energy. When this started happening hydrogen, stars lit up across our Universe.

WHERE DO ALL THE ELEMENTS COME FROM

Around 0.4 billion years after the big bang stars started to emerge all over the Universe. They burned hydrogen to create helium. Helium was used to create carbon. Neon, oxygen, silicon, and iron were also created during the lives of stars.

However, once these stars started running out of fuel is when things really got interesting. It's in the massive explosions called supernova that resulted from certain stars running out of fuel that all of the elements of the periodic table were created through neutron capture. Without the death of stars, our world would not exist today.

MOST ABUNDANT ELEMENTS IN THE MILKY WAY GALAXY IN PPM

1 Proton	^{1}H	Hydrogen	739,000
2 Protons	^{4}He	Helium	240,000
8 Protons	^{16}O	Oxygen	10,400
6 Protons	^{12}C	Carbon	4,600
10 Protons	^{20}Ne	Neon	1,340
26 Protons	^{56}Fe	Iron	1,090
7 Protons	^{14}N	Nitrogen	960
14 Protons	^{28}Si	Silicon	650
12 Protons	^{24}Mg	Magnesium	580
16 Protons	^{32}S	Sulfur	440

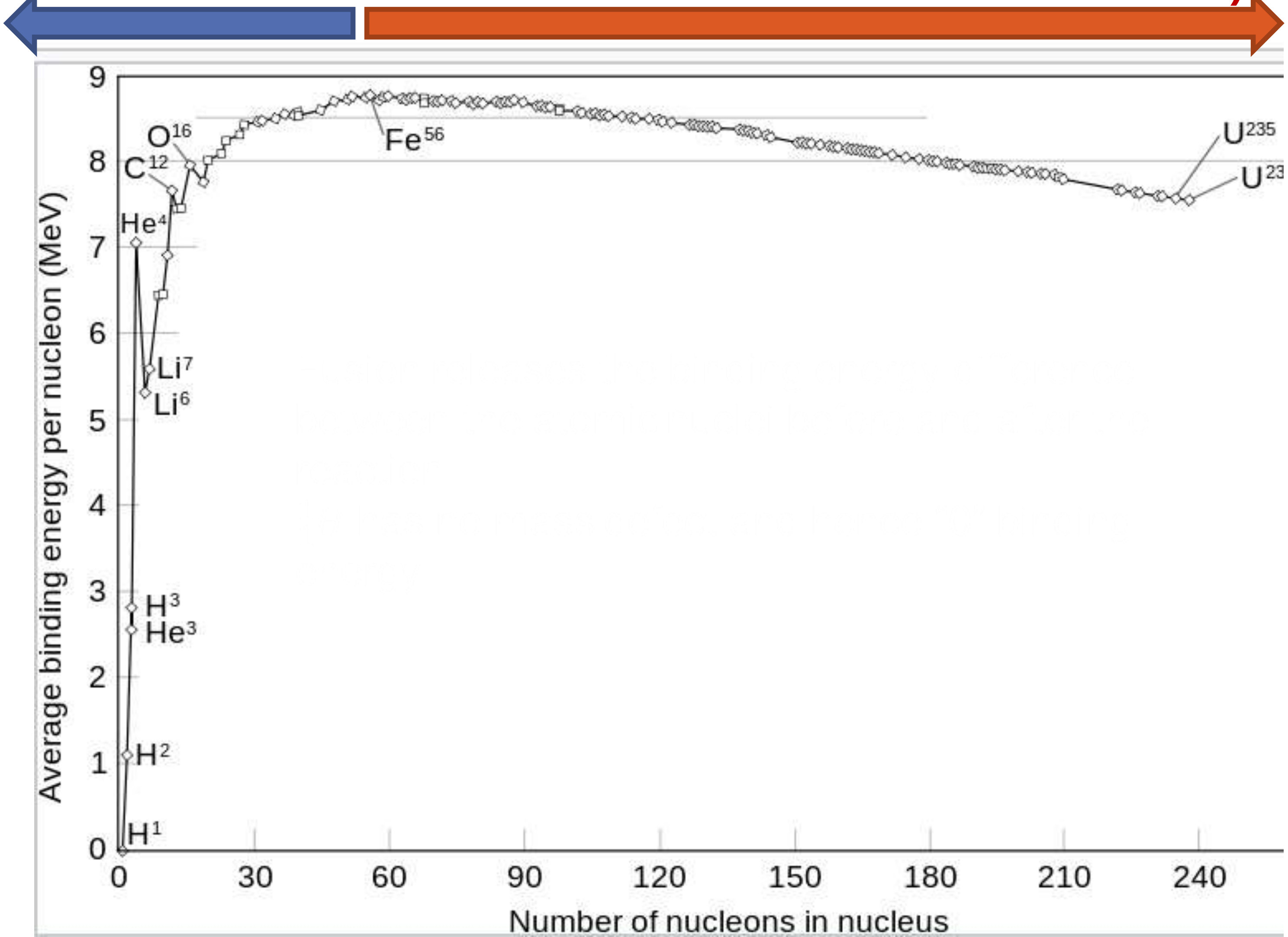
Only 2 % of the matter in the universe is more complex than hydrogen and helium

Throughout the Universe (and in our solar system) Hydrogen and Helium are most abundant (Just mostly concentrated in the sun)

HOW STARS CAN GENERATE ENERGY FROM NUCLEAR FUSION

Energy can be created through fusion

Energy can be created through fission (We do this for Uranium in a nuclear reactor)



HEAVIER ELEMENTS ARE GENERATED IN MASSIVE STARS

ELEMENTS AND ISOTOPE

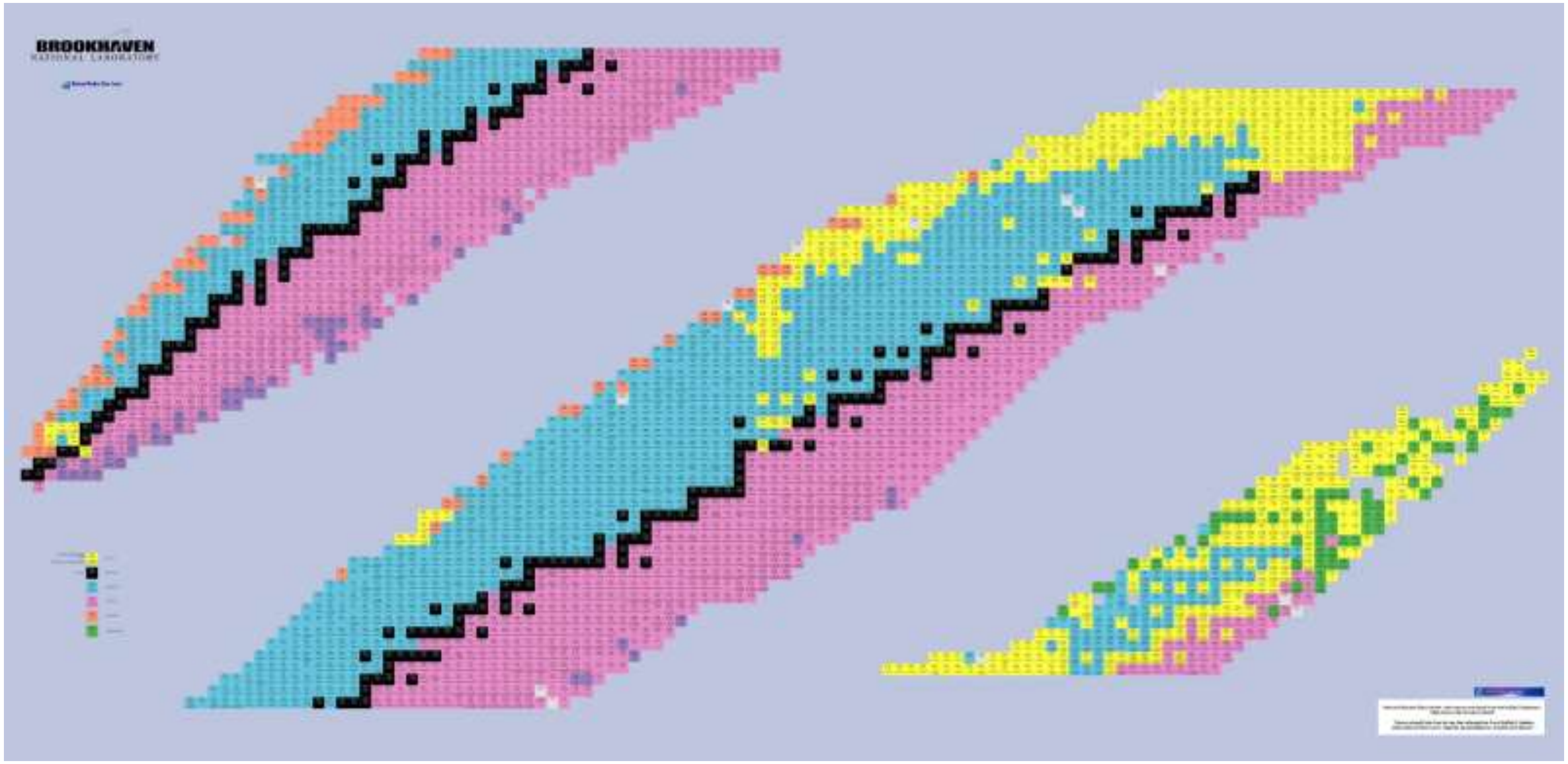
Elements: When two nuclei have different number of protons they belong to different elements. E.g. ^{14}C is an isotope of carbon while ^{14}N is an isotope of nitrogen. One has 6 protons and the other one has 7 protons.

Isotopes: When two nuclei have the same number of protons but different number of neutrons then they are called isotopes. ^{16}O, ^{17}O and ^{18}O are all stable isotopes of oxygen. All of them have 8 protons but they have 8, 9 and 10 neutrons

Radioactive decay: The vast majority of nuclei formed via fusion or neutron capture are unstable. The decay by releasing alpha, beta or gamma radiation till the nucleus reaches a stable configuration. Radioactive elements and their stable daughters (the decay products that remain behind in rocks and meteorites) tell us a lot about the early solar system and the history of Earth.

ELEMENTS AND ISOTOPE

A table of nucleotides shows all the nuclei that have ever been observed in nature or created in the lab via processes like neutron capture and it provides information on how things decay. Only the black boxes indicate stable isotopes! Stability is the exception not the rule. When you are looking for potential new evidence, potential proxy for Earth system processes or processes in the early solar system or other stars in Universe, you need to meditate in front of this chart.



ORIGIN OF SOLAR SYSTEM

Pick a theory, any theory, but it must be consistent with these facts:

1) **Planets all revolve around the Sun in the same direction in nearly circular orbits.**

2) **The angle between the axis of rotation and the plane of orbit is small (except Uranus).**

3) **All planets (except Venus and Uranus) rotate in the same direction as their revolution; their moons do, too.**

ORIGIN OF SOLAR SYSTEM

4) Each planet is roughly twice as far as the next inner planet is from the Sun (the Titus-Bode rule).

5) 99.9 % of mass is in the Sun; 99 % of angular momentum is in the planets.

6) Planets in two groups:

terrestrial (inner): Mercury, Venus, Earth, Mars contain heavy elements (O, Si, Fe, Mg) Mercury is mostly Fe (ρ = 5.4)

Jovian (outer): Jupiter, Saturn, Uranus, Neptune. Jupiter mostly gas and ice (ρ = 0.7) Pluto and other trans Neptune bodies ???

99.9% of the mass and almost all hydrogen and helium in the solar system is in the sun
The material left behind in the disk, which formed the planets mostly comprises of elements other than hydrogen and helium

SUPERNOVA SHOCK WAVE HYPOTHESIS

A shock wave from an exploding supernova injected material from the exploding star into a neighboring cloud of dust and gas, causing it to collapse in on itself and form the Sun and its surrounding planets.

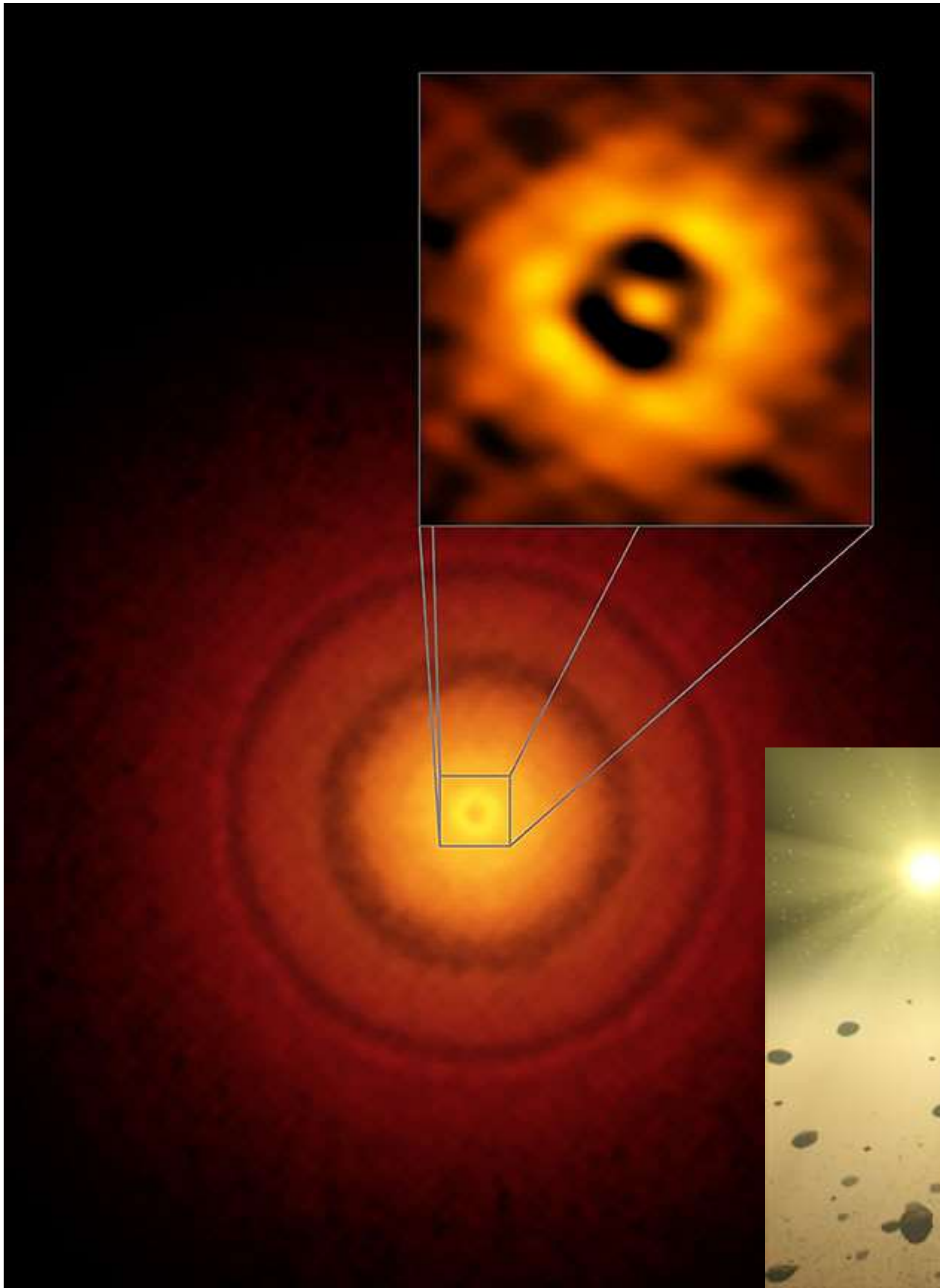
This is supported by the large amount of nickel 60 (a decay product of short lived unstable isotope iron 60) that has been found in some primitive rocky meteorites called chondrites. Iron 60 has a “short” half-life 2.6 million years

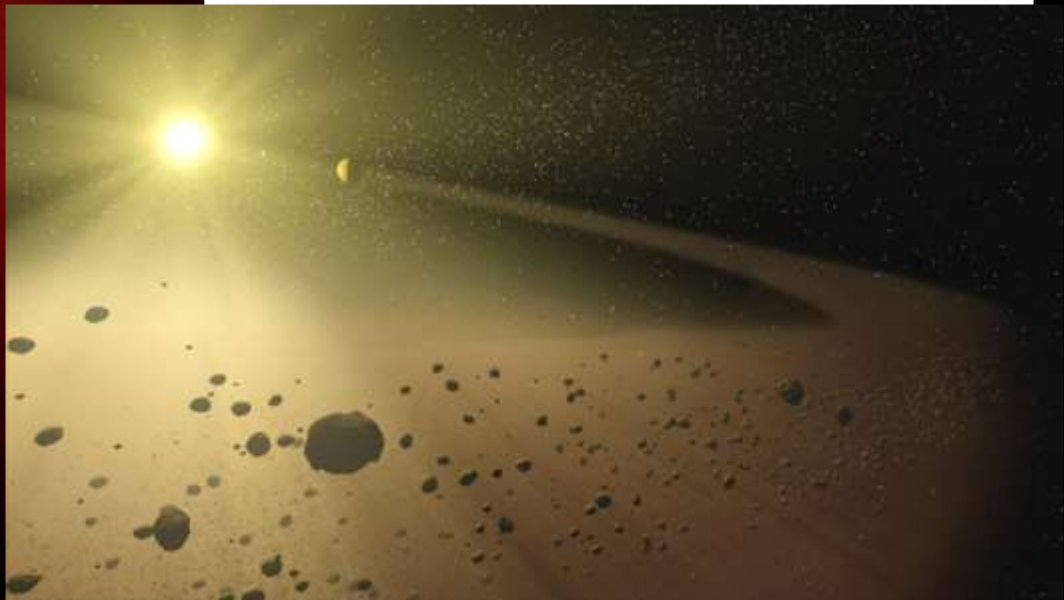
Iron 60 is only created in significant amounts by nuclear reactions inside certain kinds of stars, including in the supernovae or what are called asymptotic giant branch (AGB) stars.

A supernova shock wave tearing through the solar nebula is one of the few ways to explain how iron 60 and pre-solar grains which are isotopically very distinct from other material within the solar system could have gotten into the solar system at the time of its formation. Pre-solar grains are typically also found in chondrite meteorites.

STAGES OF THE SOLAR SYSTEM FORMATION

1. A diffuse slowly rotating nebula begins to contract under the impact of its own gravity after being perturbed by a supernova shock wave
2. As a result of the contraction a rapidly rotating flat disk forms with most of the mass (99%), particularly almost all the H, and He concentrated in the center. Heavier elements take longer to accelerate and stayed in the disk. So did some of the H, He that was further away from the center.
3. The enveloping disk of gas and dust collide and forms grains that clump together to form smaller planetesimals. Primitive meteorites called chondrites preserve some of the material that clumped together at this stage. These meteorites are used to date the solar system.
4. Terrestrial planets grew by multiple collisions and accretion of planetesimals by gravitational attraction. The giant outer planets grew by gas accretion.
5. When enough mass had concentrated in the center of the cloud fusion started and the sun lit up. Solar winds blew away the remaining H, He and other gasses from the inner terrestrial planets & Earth lost it’s first atmosphere





HOW DO WE KNOW?

A protoplanetary disk around a young sun

This picture of the nearby young star TW Hydrae reveals the classic rings and gaps that signify planets are in formation in this system. Credit: S. Andrews (Harvard-Smithsonian CfA); B. Saxton (NRAO/AUI/NSF); ALMA (ESO/NAOJ/NRAO)

WE DATE THE SOLAR SYSTEM WITH CHONDRITES

- Chondrites are primitive meteorites that represent the original material that aggregated from the dust cloud.

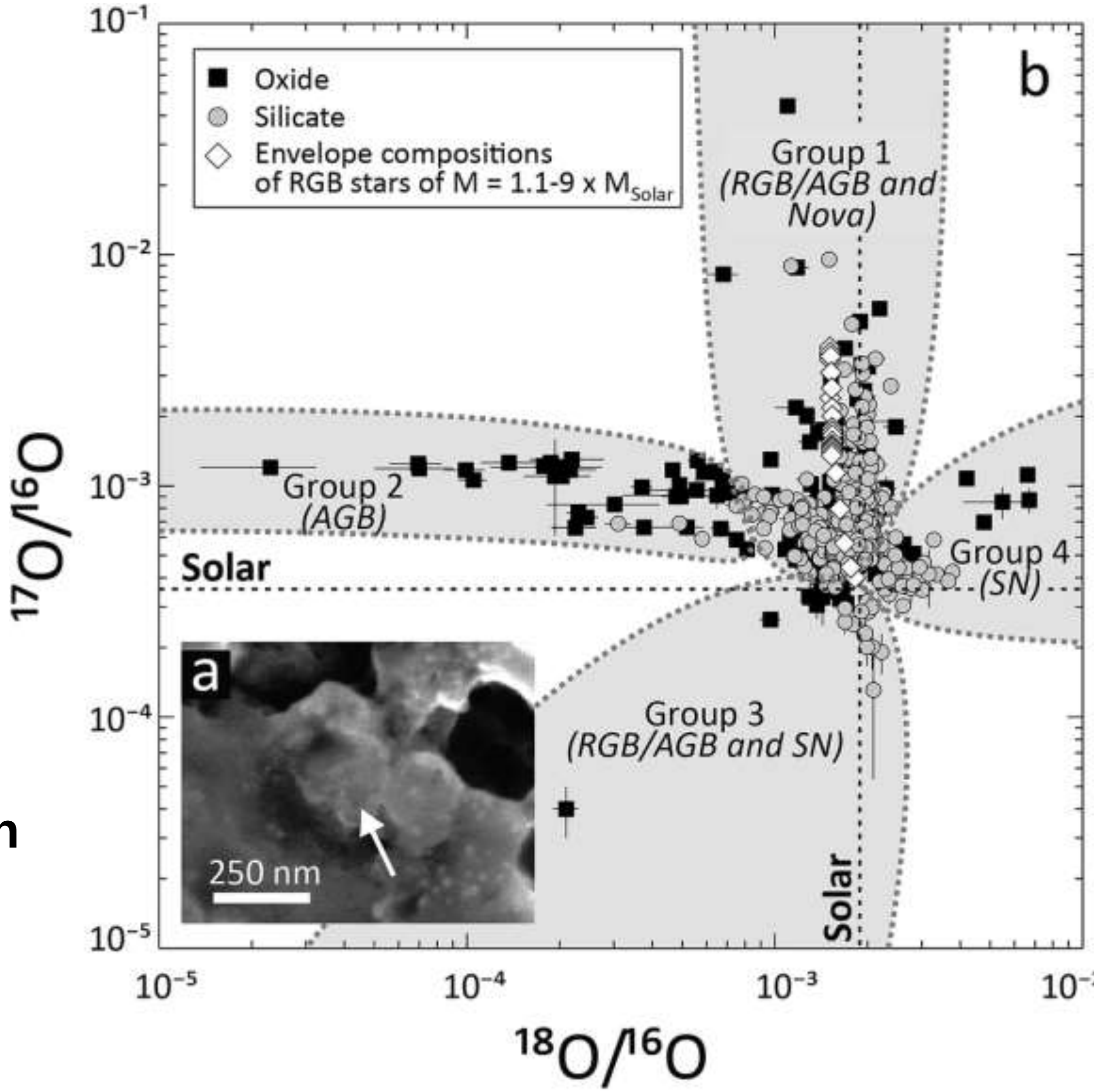
- These have never been heated enough to be melted, come from small bodies and do contain traces in the form of dust grain originating from solar systems other than our own in the form of "pre-solar grains"

The calcium–aluminium-rich inclusion or Ca–Al-rich inclusion (CAI) found in carbonaceous chondrite meteorites are probably the oldest substances in the Solar System. They are thought to have formed as fine-grained condensates from a high temperature (>1300 K) gas that existed in the protoplanetary disk . The oldest age was measured in an inclusion of the CV3 carbonaceous chondrite Northwest Africa (NWA) 2364 and was dated at 4568.22 ± 0.17 Mya using Pb-Pb dating.

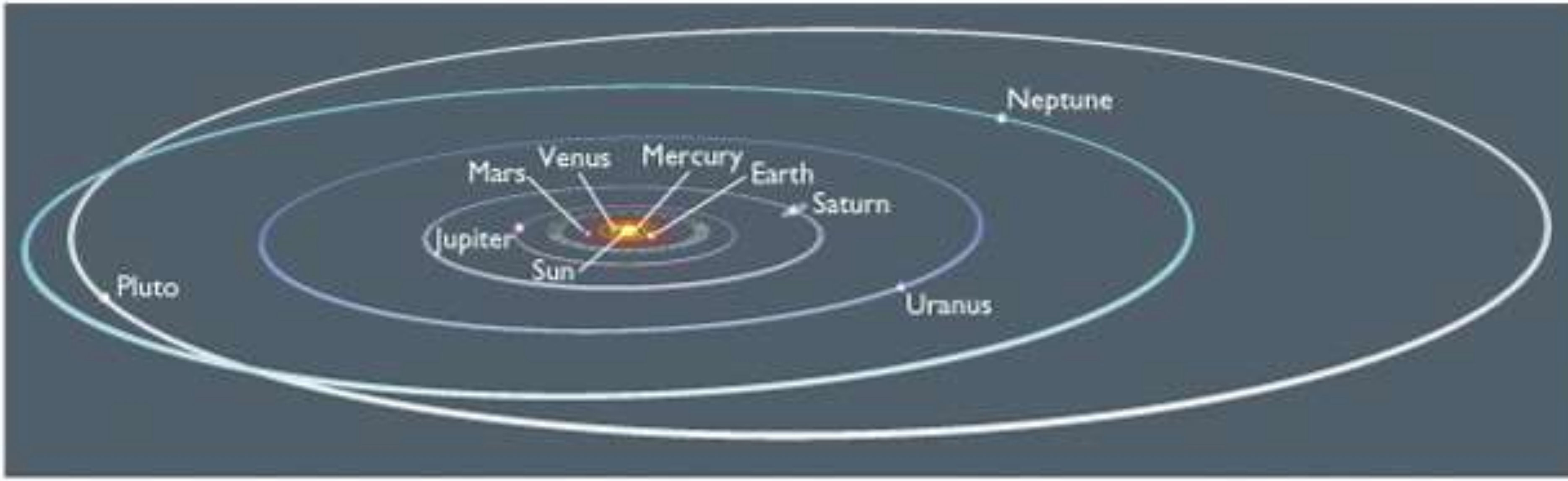
METEORITES BRING THE STORY OF OTHER STARS AND ACCRETION TO EARTH

Primitive Chondrite with presolar grain

Presolar grains formed in stars very different from our own and have very exotic isotope ratios



OUR SOLAR SYSTEM



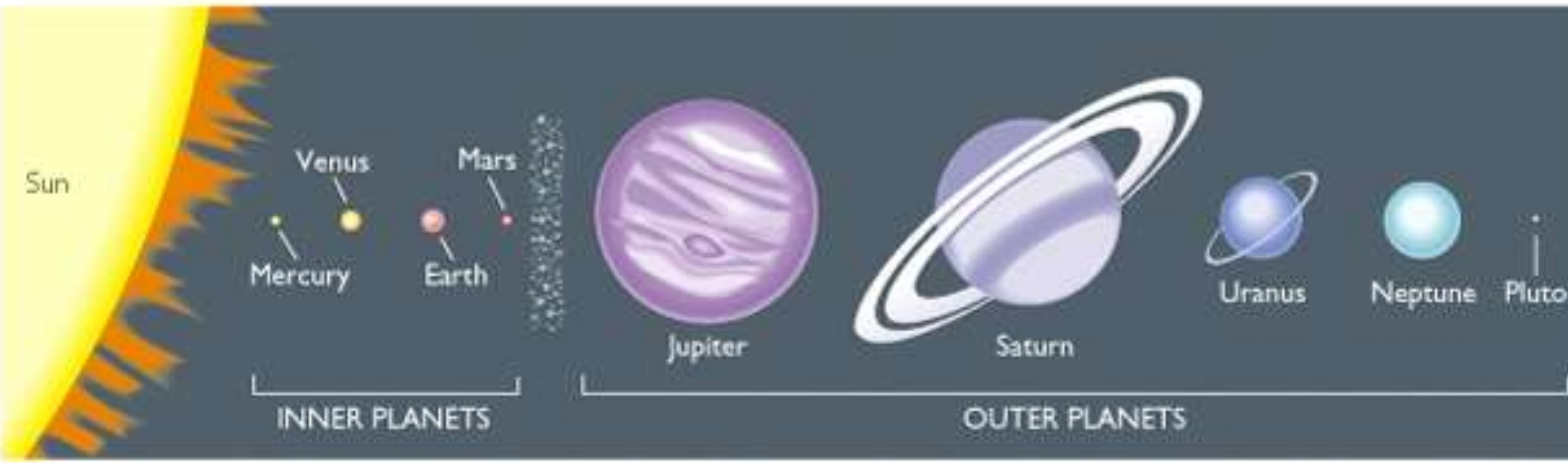


Fig. 1.3

TAKE HOME MESSAGE

Our Universe is roughly 13.8 billion years old and started with a big bang. This can be mathematically derived from the theory of relativity.

The date is experimentally derived from the rate of the expansion of the Universe which can be observed via the red shift of the light from far away galaxies.

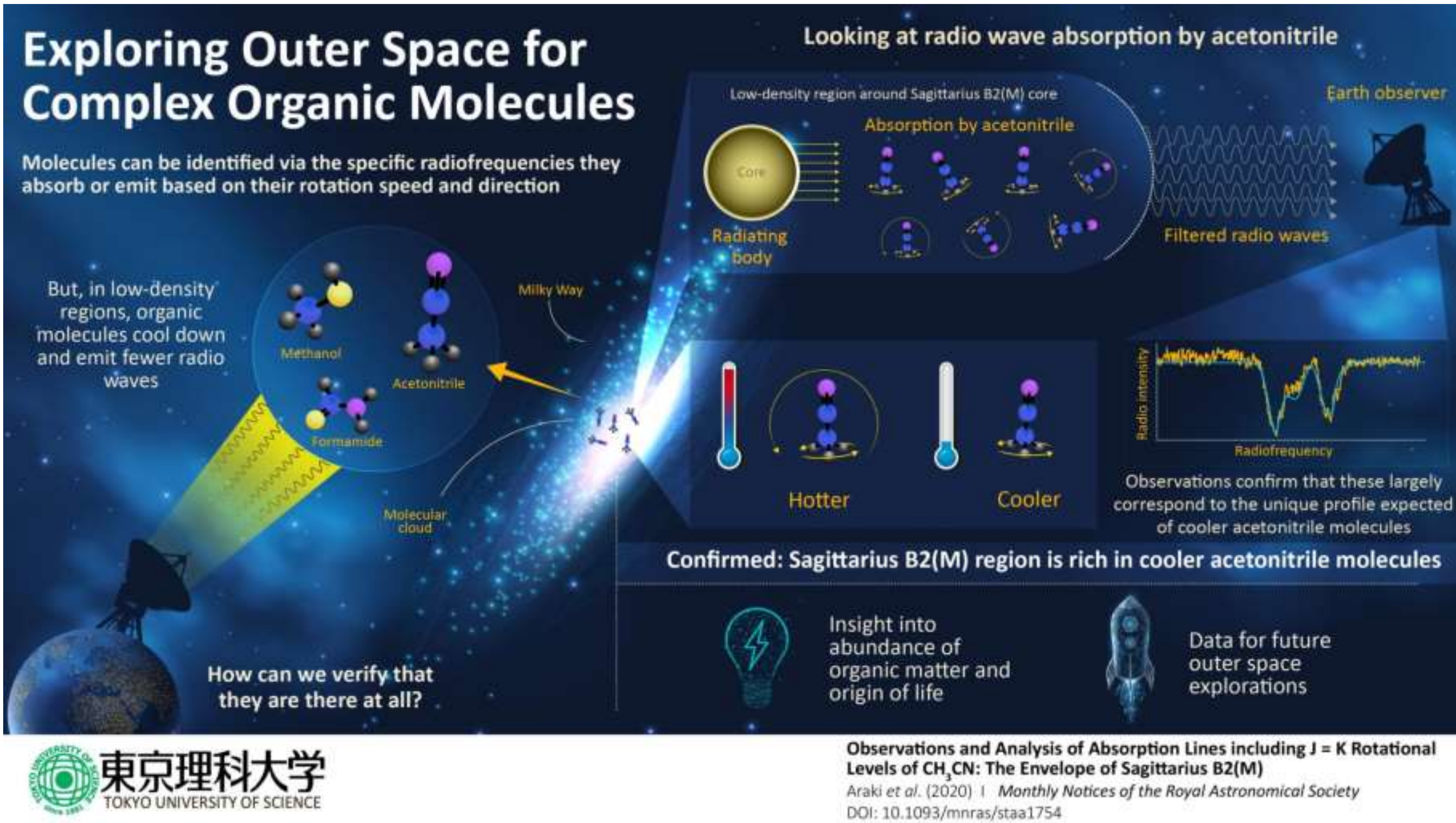
The oldest evidence we have of the early universe is the cosmic background radiation.

Our solar system formed 4.6 billion years ago. We derive the date by dating meteorites that preserve the original material which aggregated. These primitive meteorites are called chondrites.

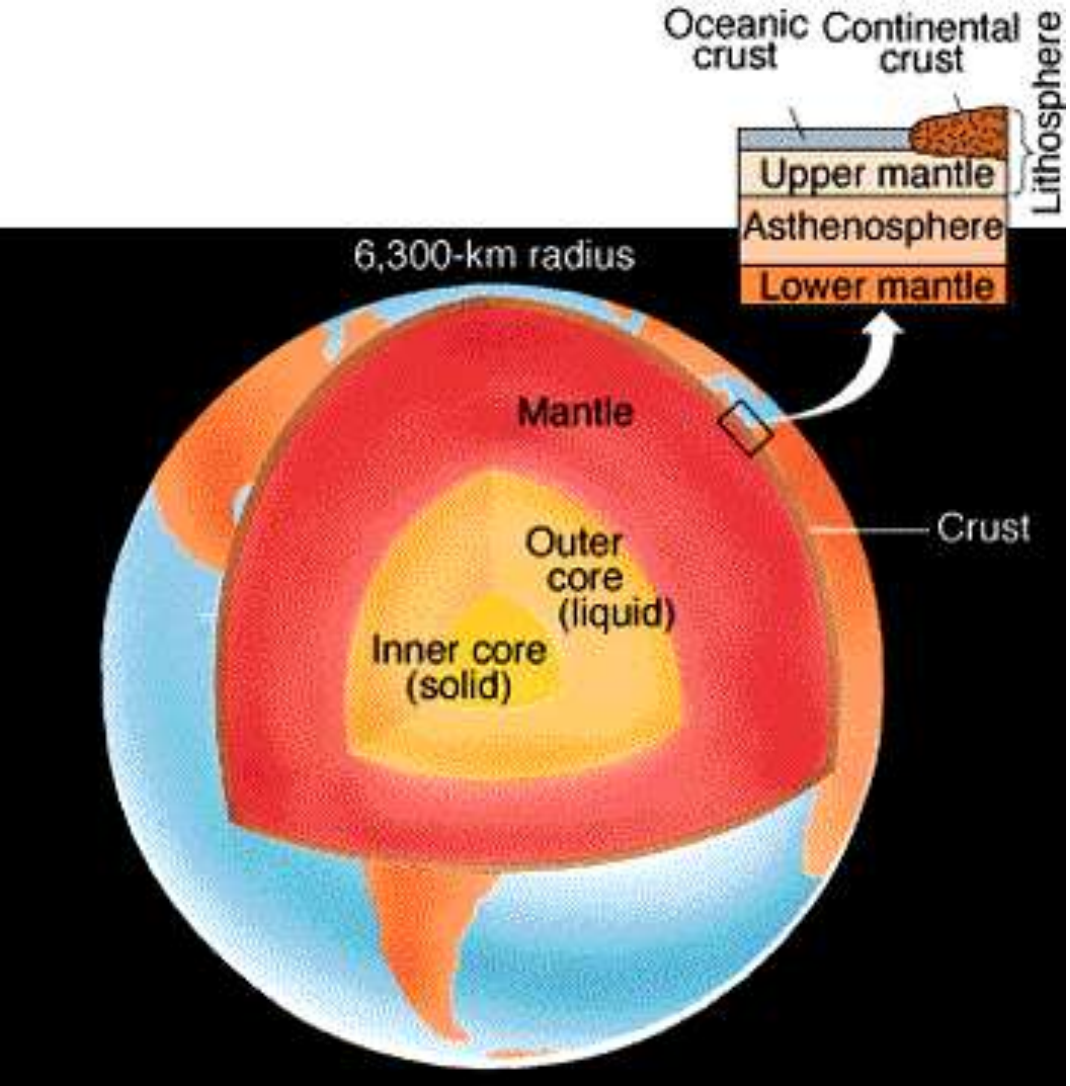
Most of our empirical evidence about processes in stars comes from disciplines such as high energy physics, nuclear physics, astrophysics, other stars that we study with spectroscopy and from tiny grains called presolar grains which reach us via chondritic meteorites and which preserve isotope ratios that could have only been formed under very different conditions in stars that died long ago.

ASTROBIOLOGY & THE MATTER PLANETS ARE MADE UP OF:

complex molecules, from carbon monoxide to polycyclic aromatic hydrocarbons, are detected in star systems



THE EARTH'S INTERNAL STRUCTURE



The Earth comprises of a crust which can be subdivided into the denser oceanic crust and the lighter continental crust, the mantle which is similar to the oceanic crust in composition but molten and a liquid outer and solid inner iron core

THE EARTH'S INTERNAL STRUCTURE

When was the hypothesis of an layered Earths core first proposed?

At the end of the 19th century by Emil Wiechert, a German physicist

The density of the earth was determined based on Newton's laws of gravity and measurements of the earths gravitational force in the 18th century as 5.5 g/cm^3

Most common rocks have a density of 3 g/cm^3 or less Wiechert calculated the effect of pressure on the density of rocks and found it is far too small to account for the difference

He knew meteorites are pieces of the solar system which fell on earth and that some where made of stone, while others were made of an alloy of iron and nickel with a density of 8 g/cm^3

He proposed the hypothesis that both types contributed to the formation of the Earth and that the heavy iron and nickel had been pulled towards the Earth's centre by the gravitational force

The hypothesis of the layered earth was tested using seismic waves

(a)

(b)

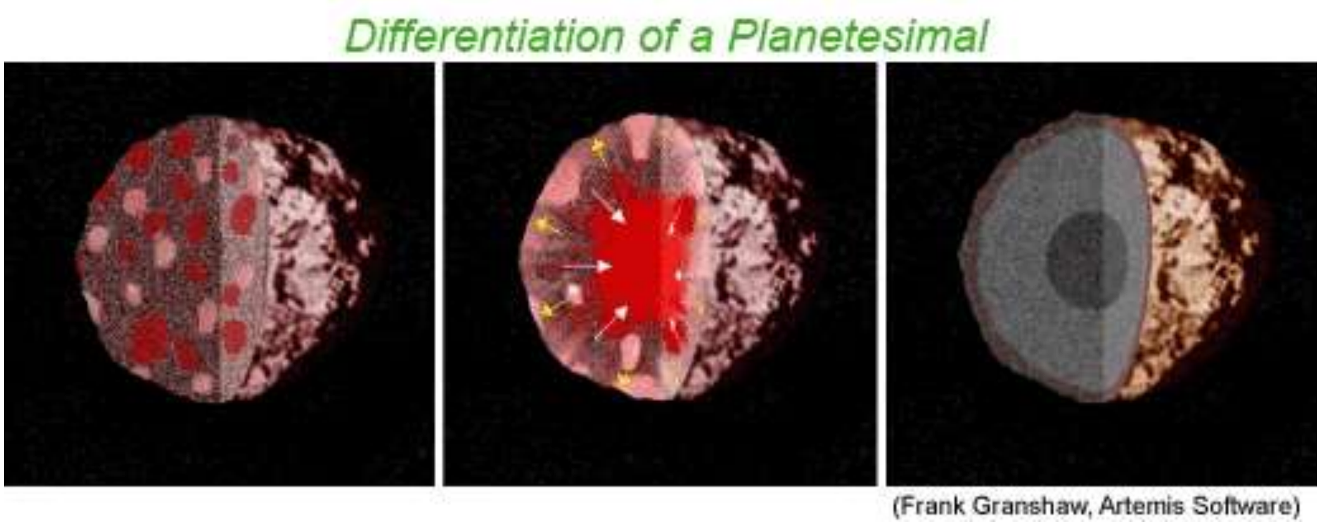
Figure 1.6 Two common types of meteorites. (a) A stony meteorite similar in composition to Earth's silicate mantle; (b) an iron-nickel meteorite similar in composition to Earth's core. The former has a density of about 3 g/cm^3, the latter a density of about 8 g/cm^3. [John Grotzinger/Ramón Rivera-Moret/ Harvard Mineralogical Museum.]

Starting material Chondrites

THE BIGEST QUESTION AFTER WIECHERT

- **If Wiechert was correct then how did an aggregation of solid junks of iron meteorites, stony meteorites and carbonaceous meteorites become a differentiated planet?**
- **Planetary differentiation was possible only if the planet had melted to at least the state of a weak solid, to allow for some diffusion/convection and had a heat source in the interior**
- **The kinetic energy released during aggregation alone was calculated to be insufficient to melt early Earth very soon after Wiechert proposed his theory. Therefore, many people did not believe Wiechert's theory. He could not account for the energy required to produce a differentiated Planet**
- **Till we developed mass spectrometers and discovered isotopes and radioactivity no one was able to explain where the energy for melting come from.**
- **Only since the proposition of the supernova shock wave hypothesis do the numbers really add up. The radioactive decay of short lived isotopes mainly ^{26}Al and ^{60}Fe provided the energy that melted early Earth.**

THE SOURCE OF THE ENERGY FOR THE INITIAL MELTING 26AL



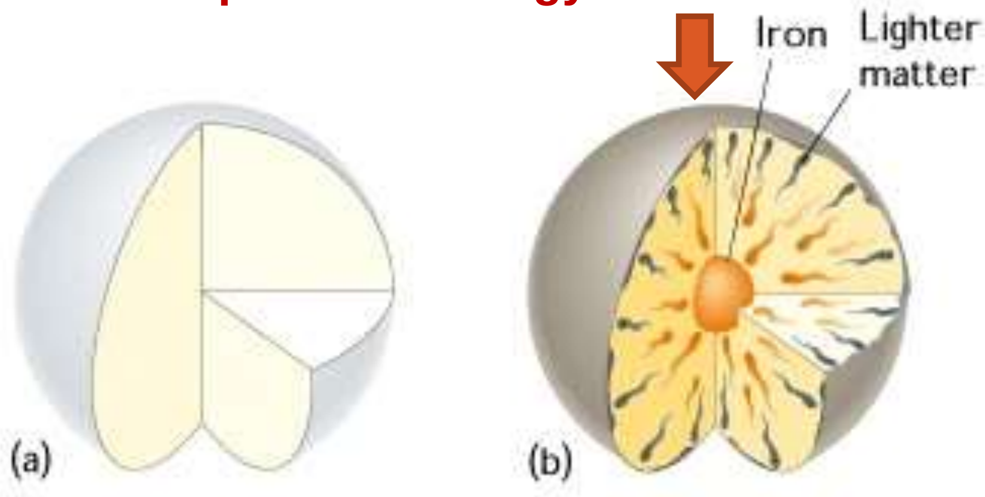
Concentrations of excess ^{26}Mg, the decay product of the short-lived radionuclide ^{26}Al (mean life = 1.03 Myr), show that the solar system formed with $n(^{26}\text{Al})/n(^{27}\text{Al}) = 5.2 \times 10^{-5}$

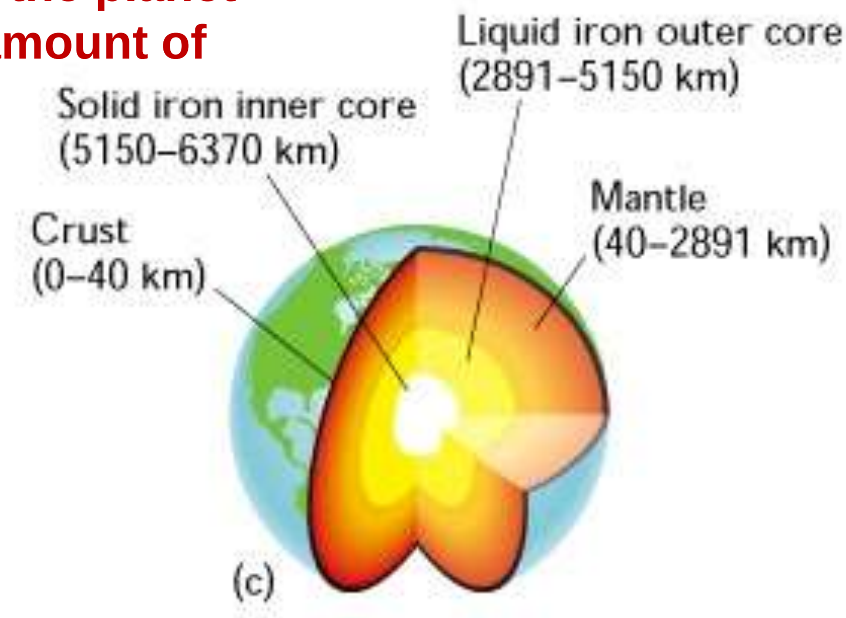
The overall concentration of ^{26}Al in the solar disk was more than a factor of 10 greater than the current average value in the interstellar medium of 3.0×10^{-6} There must have been a significant external source of this short-lived nuclide. Commonly, the natal ^{26}Al is taken as a signature of a nearby supernova that may have triggered the collapse of the molecular cloud from which the Sun formed.

A DIFFERENTIATING PLANET

At this point everything is molten & the planet covered by a magma ocean. Huge amount of potential energy is released

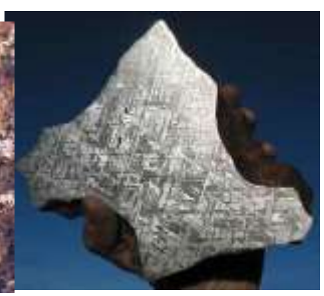
The end product





Chondritic material Is oldest

Basaltic achondrites & iron meteorites Almost the same age -> The "iron catastrophe" happened really fast



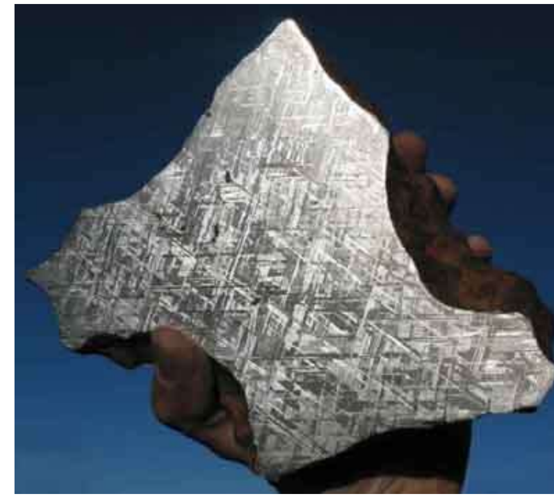
Differentiation is an ongoing process and it releases potential energy it is an internal heat engine within earth.

Fig. 1.6

ROCKY BODIES DIFFERENTIATE

Basaltic achondrites

Iron meteorite



Basaltic achondrites and iron meteorites are two sides of the same coin. We find them because some planetesimals that were large enough to melt and differentiate were thereafter broken into pieces during subsequent collisions and those pieces reached Earth.

Rocky bodies differentiate into:

rocky crusts and mantles that are depleted in **siderophiles** and enriched in **lithophiles.**

Metallic cores enriched in **siderophiles** and depleted in **lithophiles**.